Containerization-and-DevOps

🐳 DOCKER – Class Task

🛠️ Technologies Used

- Docker
- WSL (Windows Subsystem for Linux)
- Linux Commands

⚙️ Steps Performed

1️⃣ Stop Docker Service

```
sudo service docker stop
```

2️⃣ Start Docker Service

```
sudo service docker start
```

3️⃣ Kill Running Docker Daemon

```
sudo pkill dockerd
```

4️⃣ Attempt Manual Start of Docker Daemon

```
sudo dockerd &
```

5️⃣ Check Port Usage

```
ss -lntp | grep 2375
```

❌ Issue Encountered

While starting Docker daemon manually, the following error occurred:

```
failed to start daemon, ensure docker is not running or delete /var/run/docker.pid: proces
```

🔍 Cause

- Docker daemon was already running in the background
- Existing PID file (`/var/run/docker.pid`) prevented a new instance from starting

🛠️ Resolution

Option 1: Remove PID File

```
sudo rm /var/run/docker.pid
```

Option 2: Ensure No Running Docker Process

```
ps aux | grep dockerd
```

Kill if needed:

```
sudo kill -9 <PID>
```

Restart Docker Properly

```
sudo systemctl restart docker
```

⚠️ Notes

- Only one Docker daemon instance can run at a time
- Manual `dockerd` start may conflict with system service
- Always check running processes before restarting

✅ Outcome

- Identified Docker daemon conflict issue
- Understood PID file role in process management
- Applied troubleshooting steps to resolve daemon startup failure

📸 Screenshots

> screenshots below:

```
sristee@Sristee:/mnt/c/Users/srist$ sudo service docker stop
Stopping 'docker.service', but its triggering units are still active:
docker.socket
sristee@Sristee:/mnt/c/Users/srist$ sudo service docker start
sristee@Sristee:/mnt/c/Users/srist$ sudo pkill dockerd
sristee@Sristee:/mnt/c/Users/srist$ sudo dockerd &
[1] 3754
sristee@Sristee:/mnt/c/Users/srist$ INFO[2026-02-04T06:24:09.543210848Z] Starting up
failed to start daemon, ensure docker is not running or delete /var/run/docker.pid: process with PID 3519 is still running

[1]+  Exit 1                  sudo dockerd
sristee@Sristee:/mnt/c/Users/srist$ sudo dockerd &
[1] 3766
sristee@Sristee:/mnt/c/Users/srist$ INFO[2026-02-04T06:24:16.671905428Z] Starting up
failed to start daemon, ensure docker is not running or delete /var/run/docker.pid: process with PID 3519 is still running

[1]+  Exit 1                  sudo dockerd
sristee@Sristee:/mnt/c/Users/srist$ ss -lntp | grep 2375
sristee@Sristee:/mnt/c/Users/srist$ sudo dockerd &
[1] 3783
sristee@Sristee:/mnt/c/Users/srist$ INFO[2026-02-04T06:24:35.198869509Z] Starting up
failed to start daemon, ensure docker is not running or delete /var/run/docker.pid: process with PID 3519 is still running
```